

AD-CISPO: Saliency-Budgeted Clipped Importance Sampling for Reasoning RL

Abstract

Long-chain reasoning reinforcement learning depends on stable policy-gradient updates over token sequences, but current clipped objectives usually apply the same clipping radius to every generated token. This proposal studies AD-CISPO, a token-adaptive extension of MiniMax’s clipped importance-sampling policy optimization (CISPO). The central hypothesis is that CISPO’s dense token-gradient property can be preserved while improving sample efficiency by allocating a wider upper importance-sampling corridor to tokens that are estimated to be important for the sampled trajectory. The proposed novelty is a mean-preserving, saliency-budgeted upper clipping rule: the average clipping budget remains equal to the scalar CISPO baseline, but the budget is redistributed across generated tokens. After observing attention-sink failures in future-attention in-degree saliency, the next-generation extractor uses causal-tangent saliency: a first-order estimate of how much each token-position representation affects the advantage-weighted policy-gradient objective.

1 Problem and Motivation

Reinforcement learning has become a practical way to elicit mathematical and long-form reasoning from language models, especially when correctness can be checked by rule-based or programmatic rewards. PPO-style RLHF established clipped policy-gradient training as a scalable alignment method [1, 2], while GRPO removed the critic by using group-relative advantages and has been widely used for mathematical reasoning [3, 4]. More recent systems such as DAPO and MiniMax-M1 focus on stabilizing long-chain reasoning RL at scale [5, 6].

The unresolved problem is that scalar clipping treats every generated token as equally deserving of update freedom. In chain-of-thought-style reasoning, this is unlikely to be true. A token that introduces an intermediate quantity, selects an operation, or anchors a later conclusion may matter more than a formatting token or locally redundant phrase. A single global clipping radius therefore creates an undesirable tradeoff:

- if the radius is tight, important reasoning tokens may be prevented from moving enough when the reward signal identifies a better trajectory;
- if the radius is loose, low-value or unstable tokens receive the same update corridor, increasing variance and drift;
- if PPO/GRPO surrogate clipping suppresses token updates, the model can lose useful gradient signal exactly when the sampled policy has moved outside the trust corridor.

CISPO directly targets the third issue by clipping the importance-sampling (IS) weight itself and using the detached clipped weight to scale $\log \pi_\theta$, so clipped tokens still contribute gradients [6]. AD-CISPO asks whether the remaining scalar-budget issue can be addressed without abandoning CISPO’s dense-gradient design.

2 Existing Solutions and Limitations

PPO and GRPO. PPO clips the policy-ratio surrogate to control policy updates while allowing multiple optimization epochs per batch [1]. GRPO adapts this family of methods to reasoning models by computing advantages from a group of sampled responses rather than a learned critic [3]. These methods are simple and effective, but their token-level clipping rules still use fixed trust-region hyperparameters.

DAPO and asymmetric clipping. DAPO improves long-chain reasoning RL with techniques such as a higher upper clip, dynamic sampling, token-level policy-gradient loss, and overlong reward shaping [5]. This is an important baseline because it shows that the upper clipping corridor matters for reasoning RL. However, DAPO still treats the upper corridor as a sequence-wide or globally configured scalar, not as a budget to be distributed across tokens.

CISPO. MiniMax-M1 proposes CISPO, which clips IS weights rather than clipping away token updates and reports improved RL efficiency in large-scale reasoning training [6]. CISPO is the strongest starting point for this proposal because it preserves per-token gradient density. Its limitation for this project is that the upper IS clipping radius remains scalar.

Attention-based saliency. Transformers route information through attention mechanisms [7], and attention patterns can provide useful structural signals. However, attention weights should not be treated as direct causal explanations of model behavior [8]. Attention sinks are a concrete example: initial or special tokens may receive high attention for non-semantic reasons [9]. AD-CISPO therefore uses frozen-reference attention only as an update-budget heuristic, not as a claim about human-interpretable explanation, and explicitly masks prompt and sink tokens.

3 Research Gap

Existing reasoning-RL objectives answer two separate questions only partially:

1. How can policy-gradient updates remain stable under off-policy drift? PPO, GRPO, DAPO, and CISPO address this through clipped ratios or clipped IS weights.
2. Which generated tokens deserve a larger or smaller update corridor? Existing methods largely do not answer this; they apply scalar clipping budgets across all tokens.

The gap is a token-budgeted trust-region objective for language-model RL: one that keeps the same average clipping budget as a scalar baseline, but reallocates that budget according to a token-importance signal tied to the sampled trajectory. AD-CISPO is designed to fill this gap for CISPO-style clipped IS-weight policy gradients.

4 Proposed Solution

AD-CISPO replaces the scalar upper IS clipping radius in CISPO with a token-adaptive upper radius. The lower bound remains inactive by default to match the MiniMax-style CISPO baseline. Each generated token receives a multiplier computed from a saliency extractor, and the multipliers are normalized so their mean over valid generated tokens is 1. This preserves the same average clipping budget as scalar CISPO while changing where the budget is spent. The first implementation

used frozen-reference future-attention in-degree; the next-generation implementation should use causal-tangent saliency because attention in-degree can be dominated by sink tokens.

Core hypothesis. For reasoning tasks with sparse sequence-level rewards, reallocating CISPO’s upper clipping budget toward structurally reused generated tokens will improve learning efficiency or stability relative to scalar CISPO, while shuffled or uniform multipliers should remove the gain if the saliency signal is doing real work.

What is novel. The proposal’s novelty is not the use of one particular saliency heuristic by itself. The novel contribution is the combination of:

1. CISPO-compatible token-adaptive upper IS clipping that preserves dense token gradients;
2. mean-preserving multiplier normalization, so AD-CISPO changes budget allocation rather than increasing the total average clipping budget;
3. an explicit saliency interface, where attention in-degree is a cheap baseline and causal-tangent saliency is the first objective-sensitive replacement;
4. sink-token diagnostics and guards, so punctuation or boilerplate attention sinks cannot silently absorb the adaptive clipping budget;
5. falsifiable ablations that separate saliency from extra variance, asymmetric clipping, and reference-model overhead.

5 Notation

Symbol	Meaning	Shape or domain
q	prompt/question	text
i	index of a sampled response within a prompt group	$i \in \{1, \dots, G\}$
o_i	response i in a sampled group	token sequence
t	token position in response i whose saliency, IS weight, or multiplier is being computed	$t \in M_i$
j	later token position in response i that attends back to position t	$j \in M_i, j > t$
u	summation index over valid generated positions in response i	$u \in M_i$
$o_{i,t}$	token at position t of response i	token id
G	number of responses sampled per prompt	positive integer
π_θ	current policy being optimized	language model
$\pi_{\theta_{\text{old}}}$	rollout policy that produced the batch	language model
π_{ref}	frozen reference model for optional KL and saliency baselines	language model
$r_{i,t}(\theta)$	IS weight $\frac{\pi_\theta(o_{i,t} q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} q, o_{i,<t})}$	positive scalar
$\hat{A}_i$	group-relative sequence advantage for response i	scalar
M_i	valid generated-token action positions for response i	token-position set
$S_{i,t}$	raw saliency for token $o_{i,t}$	nonnegative scalar
$m_{i,t}$	normalized token multiplier	nonnegative scalar, mean 1 over M_i
$\epsilon_{\text{low}}^{\text{IS}}$	lower IS clipping radius	nonnegative scalar
$\epsilon_{\text{high}}^{\text{IS}}$	scalar upper IS clipping radius	positive scalar
$\epsilon_{i,t,\text{high}}^{\text{IS}}$	token-adaptive upper IS clipping radius	positive scalar

Index conventions. For each prompt, the model samples G responses; i selects one of them. Within response i , positions t , j , and u all refer to token indices in the same sequence, but play different roles: t is the position whose saliency or clipping budget is being assigned; j is a *later* position whose attention query looks back at t ; and u is a dummy index used only when averaging or summing over all valid generated positions in M_i .

6 Baseline: MiniMax-Style CISPO

The CISPO baseline starts from the off-policy REINFORCE objective with the IS weight treated as a stop-gradient coefficient:

$$J_{\text{ISPG}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \text{sg}(r_{i,t}(\theta)) \hat{A}_i \log \pi_{\theta}(o_{i,t} \mid q, o_{i,<t}) \right]. \quad (1)$$

CISPO clips the IS weight itself:

$$\hat{r}_{i,t}(\theta) = \text{clip}(r_{i,t}(\theta), 1 - \epsilon_{\text{low}}^{IS}, 1 + \epsilon_{\text{high}}^{IS}), \quad (2)$$

and optimizes

$$J_{\text{CISPO}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \text{sg}(\hat{r}_{i,t}(\theta)) \hat{A}_i \log \pi_{\theta}(o_{i,t} \mid q, o_{i,<t}) \right]. \quad (3)$$

A faithful baseline for this proposal sets the KL coefficient to zero and avoids an active lower IS-weight bound, following the MiniMax-M1 description of CISPO [6]. This isolates the effect of the upper clipping budget.

7 AD-CISPO Objective

AD-CISPO replaces the scalar upper clipping radius with a token-adaptive upper radius:

$$\epsilon_{i,t,\text{high}}^{IS} = \epsilon_{\text{high}}^{IS} m_{i,t}. \quad (4)$$

The clipped weight becomes

$$\hat{r}_{i,t}^{\text{AD}}(\theta) = \text{clip}(r_{i,t}(\theta), 1 - \epsilon_{\text{low}}^{IS}, 1 + \epsilon_{i,t,\text{high}}^{IS}), \quad (5)$$

and the objective is

$$J_{\text{AD-CISPO}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \text{sg}(\hat{r}_{i,t}^{\text{AD}}(\theta)) \hat{A}_i \log \pi_{\theta}(o_{i,t} \mid q, o_{i,<t}) \right]. \quad (6)$$

The adaptive multiplier is computed from token saliency. Let M_i denote the set of generated-token action positions in response i that are eligible for adaptive clipping. It excludes prompt positions, padding, masked-out actions, special tokens, and any explicit sink-token categories removed by the saliency guard. Here u ranges over all positions in M_i to form the mean saliency in the denominator, while t is the position whose multiplier is being computed:

$$\tilde{m}_{i,t} = \frac{S_{i,t}}{\frac{1}{|M_i|} \sum_{u \in M_i} S_{i,u}}. \quad (7)$$

If all valid saliency values are zero or non-finite, $\tilde{m}_{i,t}$ falls back to 1 for valid tokens. If lower or upper multiplier bounds are used, they must be followed by renormalization or projection so that

$$\frac{1}{|M_i|} \sum_{t \in M_i} m_{i,t} = 1. \quad (8)$$

A plain clamp is insufficient because it changes the total clipping budget.

8 Saliency Estimation

The main AD-CISPO implementation uses exact frozen-reference future-attention in-degree. For a selected reference layer ℓ and head h , let $a_{i,j,t}^{(\ell,h)}$ denote the causal attention probability from the query at token position j to the key/value position t . Position t is the saliency target; position j is each later valid generated token that attends to t . The saliency assigned to token $o_{i,t}$ is

$$S_{i,t} = \frac{1}{|L_{\text{top}}|} \sum_{\ell \in L_{\text{top}}} \frac{1}{H_\ell} \sum_{h=1}^{H_\ell} \sum_{\substack{j \in M_i \\ j > t}} a_{i,j,t}^{(\ell,h)}. \quad (9)$$

Thus a token is salient when later valid generated tokens attend to it under the frozen reference model. The strict condition $j > t$ prevents self-attention from inflating saliency. In this estimator, M_i is used both as the source query set and target key set, so prompt, padding, special-token, and explicitly guarded sink-token positions do not participate in the saliency budget.

The implementation computes this quantity blockwise from reference-model post-RoPE query and key states. This avoids materializing a full batch of $T \times T$ attention matrices while preserving exact softmax attention weights for the selected layers, including causal, padding, and sliding-window masks.

A cheaper ablation is frozen-reference key-vector norm:

$$S_{i,t}^{\text{KV}} = \frac{1}{|L_{\text{top}}|} \sum_{\ell \in L_{\text{top}}} \left\| k_{i,t}^{(\ell)} \right\|_2. \quad (10)$$

It should be reported as a proxy baseline, not as the primary saliency definition.

9 Mathematical Failure Mode: Attention Sinks versus Causal Importance

The future-attention in-degree estimator is best understood as an attention-centrality score:

$$S_{i,t} = \sum_{\ell,h} \sum_{\substack{j \in M_i \\ j > t}} a_{i,j,t}^{(\ell,h)}. \quad (11)$$

It answers the question “how much do later generated tokens look back at position t ?” AD-CISPO, however, ultimately needs a different quantity: how much changing the representation at position t would change the policy-gradient objective. These two quantities can diverge because attention mass is not a causal explanation of model behavior [8].

In a transformer attention layer, the value mixed into query position j is

$$z_{i,j} = \sum_{t \leq j} a_{i,j,t} V_{i,t}. \quad (12)$$

A large coefficient $a_{i,j,t}$ says that position j reads from position t , but it does not imply that $V_{i,t}$ changes the final answer, reward, or policy-gradient update. This distinction matters for attention sinks [9]: early separator or formatting tokens can attract many small attention weights from later positions for non-semantic reasons.

For example, suppose a response begins with a common reasoning preamble:

Okay, let's try to figure out ...

Let t_c be the comma position. If many future layer-head-query terms assign a small amount of attention to this comma, the accumulated in-degree can dominate:

$$a_{i,j,t_c}^{(\ell,h)} \approx 0.02, \quad 400 \text{ contributing terms} \Rightarrow S_{i,t_c} \approx 8.0. \quad (13)$$

A genuinely reasoning-critical token may receive stronger attention from fewer positions:

$$20 \text{ contributing terms} \times 0.10 \Rightarrow S_{i,t_{\text{math}}} \approx 2.0. \quad (14)$$

Under attention centrality, the comma receives the larger score even though it may not be useful for solving the problem.

This creates a direct optimizer-level failure. AD-CISPO converts saliency into an upper clipping budget:

$$\epsilon_{i,t,\text{high}}^{IS} = \epsilon_{\text{high}}^{IS} m_{i,t}. \quad (15)$$

If the comma sink receives $m_{i,t_c} = 2$ and $\epsilon_{\text{high}}^{IS} = 0.2$, its upper IS bound becomes

$$1 + \epsilon_{i,t_c,\text{high}}^{IS} = 1 + 0.2 \cdot 2 = 1.4. \quad (16)$$

Thus the largest adaptive trust region can be allocated to punctuation in a boilerplate prefix rather than to the tokens whose representations influence the rewarded reasoning trajectory.

Causal-tangent saliency. A better first-generation replacement is an attribution-patching-style first-order estimator, which approximates causal intervention effects with a single backward pass [10, 11, 12]. Define a teacher-forced scalar objective over sampled completions:

$$F(\theta) = \sum_i |\hat{A}_i| \frac{\sum_{t \in M_i} \log \pi_\theta(o_{i,t} \mid q_i, o_{i,<t})}{|M_i|}. \quad (17)$$

Let $h_{i,t}^{(\ell)}$ be the residual-stream activation at selected layer ℓ and generated position t . A first-order Taylor expansion gives

$$F(h_{i,t}^{(\ell)} + \Delta) - F(h_{i,t}^{(\ell)}) \approx \nabla_{h_{i,t}^{(\ell)}} F \cdot \Delta. \quad (18)$$

The word “causal” is used in the intervention sense, not as a claim that the score is a complete causal proof. The ideal causal question is: if one could intervene on the hidden state at token position t while holding the rest of the computation fixed as much as possible, how much would the scalar objective F change? Exact activation patching answers this by replacing or perturbing $h_{i,t}^{(\ell)}$ and rerunning the downstream computation, but doing this for every token, layer, and rollout is too

expensive inside RL training. The first-order term above is the tangent-space approximation to that intervention: it estimates the local effect of an infinitesimal do-intervention $\text{do}(h_{i,t}^{(\ell)} \leftarrow h_{i,t}^{(\ell)} + \Delta)$ on F . Thus the method is “causal” because it estimates the effect of changing an internal state on the objective, and “tangent” because it uses the local linearization rather than an explicit finite intervention.

This suggests a token saliency score

$$S_{i,t}^{\text{CT}} = \sum_{\ell \in L_{\text{top}}} \frac{\left| \nabla_{h_{i,t}^{(\ell)}} F \cdot \left(h_{i,t}^{(\ell)} - \frac{1}{|M_i|} \sum_{u \in M_i} h_{i,u}^{(\ell)} \right) \right|}{\sqrt{d_{\text{model}}}}. \quad (19)$$

The geometric intuition behind this score is that $\nabla_{h_{i,t}^{(\ell)}} F$ is the normal vector to the local tangent plane of the training objective with respect to the token-position representation. The centered activation

$$\bar{h}_{i,t}^{(\ell)} = h_{i,t}^{(\ell)} - \frac{1}{|M_i|} \sum_{u \in M_i} h_{i,u}^{(\ell)} \quad (20)$$

is a direction from the response’s average generated-token state toward the specific token’s representation. Their inner product measures how much the token-specific representation points in a direction that locally increases or decreases the scalar objective. Taking the absolute value treats both positive and negative local influence as importance: a token can matter because it supports the rewarded trajectory, or because the objective is strongly trying to move away from its current representation.

This differs from attention in-degree in three important ways. First, a token must be connected to the objective gradient; receiving attention is not enough. If $\nabla_{h_{i,t}^{(\ell)}} F \approx 0$, then $S_{i,t}^{\text{CT}}$ is small even when the token is an attention sink. Second, centering removes common sequence-level activation structure, so generic boilerplate magnitude is less likely to dominate. Third, summing over the final selected layers asks whether the token representation remains useful near the model’s decision surface, rather than whether it was merely read by later tokens in an earlier attention computation.

For AD-CISPO, the practical interpretation is:

allocate more upper IS clipping budget to tokens whose hidden states lie in directions to which the actual policy-gradient objective is locally sensitive.

This is why causal-tangent saliency is a better match to adaptive clipping than attention centrality. The clip multiplier is no longer driven by how often a token is looked at; it is driven by a first-order estimate of how much the training objective would change if that token-position representation were perturbed.

The proposed next-generation extractor should therefore treat attention in-degree as a diagnostic baseline and use causal-tangent saliency, plus explicit sink-token guards, as the first serious replacement when attention sinks dominate the top-saliency table.

Target policy versus frozen policy. Causal-tangent saliency should use activations and gradients from the current target policy π_θ , not from the frozen reference policy π_{ref} . The reason is built into the definition of F : it is the teacher-forced objective whose log-probability terms are produced by the model being updated. The gradient

$$\nabla_{h_{i,t}^{(\ell)}} F \quad (21)$$

is therefore a sensitivity of the target policy’s computation graph. If the same score were computed on π_{ref} , it would estimate the effect of perturbing hidden states in a model that will not receive the CISPO update. That would reintroduce the same mismatch that made frozen-reference attention-saliency fragile: the saliency source could emphasize obsolete features or sink behavior from a policy that is no longer the object being optimized.

The frozen policy remains useful as a baseline and diagnostic. It can provide KL/reference metrics, future-attention in-degree ablations, and drift measurements. But the primary causal-tangent extractor should run after rollouts are collected and before the policy update, using the current training model π_θ on the sampled prompt-response sequences.

10 Implementation Plan

The minimum implementation should use the existing Curious training path and GSM8K reward setup [13]. The intended experimental model is a small reasoning-capable open model, such as the repository’s Qwen3-1.7B configuration, so that full ablations are affordable on a single H100.

Next-generation implementation flow: $\pi_{\theta_{\text{old}}}$ samples grouped responses $\rightarrow$ verifier computes rewards and group advantages $\rightarrow$ target policy is run teacher-forced on sampled responses $\rightarrow$ final-layer hidden states are captured $\rightarrow$ the advantage-weighted log-prob objective F is backpropagated to those hidden states $\rightarrow$ causal-tangent saliency $S_{i,t}^{\text{CT}}$ is computed and sink-guarded $\rightarrow$ saliency is normalized into $m_{i,t}$ $\rightarrow$ token thresholds $\epsilon_{i,t,\text{high}}^{\text{IS}} = \epsilon_{\text{high}}^{\text{IS}} m_{i,t}$ $\rightarrow$ CISPO loss uses detached clipped IS weights $\rightarrow$ metrics log reward, exact match, clipping, saliency shape, sink-guard behavior, and runtime overhead.

Implementation contract. A mathematically aligned AD-CISPO implementation should satisfy the following:

1. CISPO loss must use a detached clipped IS weight multiplying $\log \pi_\theta$, not a PPO/GRPO surrogate minimum.
2. AD-CISPO must be enabled only with the CISPO loss, unless explicitly naming and evaluating an adaptive-GRPO ablation.
3. MiniMax-style CISPO should set KL weight to zero and avoid an active lower IS-weight bound.
4. Token-adaptive clipping should change only the upper IS clipping radius unless a lower-bound ablation is deliberately introduced.
5. Saliency tensors should align with the generated token whose IS weight is being clipped. In next-token prediction implementations, generated-token saliency must align with next-token log-probability terms, not the preceding hidden-state position.
6. Causal-tangent saliency should be computed from the target policy on the sampled completions, with the scalar objective using the same generated-token log-probability terms and group advantages that drive the CISPO update. Frozen-policy activations should be reserved for explicit ablations, not for the primary extractor.
7. The causal-tangent backward pass should differentiate only the saliency objective used to score token positions; the final CISPO update should still use the normal training loss and should not accidentally reuse stale gradients from saliency extraction.

8. Sink guards should be explicit and logged. Separator punctuation, whitespace, special tokens, and prompt tokens should either be removed from multiplier assignment or capped at multiplier 1, while numeric tokens and mathematical operators should remain eligible unless ablated.
9. Claims about compute and memory must distinguish active-policy saliency overhead from frozen-reference overhead.

Causal-tangent extractor steps. The recommended first implementation is:

1. Add a saliency method flag such as `ad_cispo_saliency_method=causal_tangent`.
2. During trajectory processing, run the target policy teacher-forced over the sampled prompt-response sequences, using the same logits-minibatch boundary used for log-probability computation. This should use the current training model before the update step, not the frozen reference model.
3. Register hooks on the last four decoder layers and retain gradients for their hidden-state tensors at generated-token positions.
4. Form F from generated-token log-probabilities weighted by $|\hat{A}_i|$, with per-response length normalization so long completions do not dominate saliency solely by length.
5. Backpropagate F , compute $S_{i,t}^{\text{CT}}$ from gradient-centered-activation inner products, then detach and clear temporary gradients before the policy update.
6. Apply sink guards and log the fraction of tokens and saliency mass removed by those guards.
7. Reuse the existing mean-preserving multiplier projection and AD-CISPO clip-high plumbing.

11 Evaluation Plan

Datasets and tasks. The evaluation should use a staged math-reasoning suite rather than a single benchmark:

1. **GSM8K** should remain the first target, because it has a clear exact-answer reward and is already supported in Curious [13]. It is the right dataset for fast debugging, seed sweeps, and early reward-curve comparisons.
2. **SVAMP** should be used as an evaluation-only robustness check for arithmetic word-problem variations [14]. This helps test whether AD-CISPO improves reasoning robustness rather than merely fitting GSM8K surface patterns.
3. **MATH and MATH-500** should be the next difficulty tier. MATH provides competition-style problems with step-by-step solutions [15], while MATH-500 is a smaller representative subset commonly used for cheaper repeated evaluation [16]. These tasks test whether the token-adaptive clipping signal transfers to longer derivations and more diverse mathematical domains.
4. **AIME/AMC-style held-out sets** should be reserved for final stress testing once answer extraction is stable. If broader olympiad-level coverage is needed, add the mathematics split of OlympiadBench as an optional evaluation-only benchmark [17].

Training should begin with GSM8K and expand only after scalar CISPO and AD-CISPO have matched data loading, answer extraction, rollout count, and compute budgets. Harder datasets should be used primarily to measure transfer, not to tune the AD-CISPO saliency heuristic.

Baselines. The minimum comparison set is:

1. GRPO/PPO-style surrogate baseline.
2. DAPO-style asymmetric clipping or clip-high baseline.
3. MiniMax-style CISPO: no KL, no active lower IS bound, scalar upper IS radius.
4. AD-CISPO with causal-tangent saliency and sink guards.
5. AD-CISPO with exact future-attention in-degree saliency.
6. AD-CISPO with KV-norm proxy saliency.
7. Shuffled-saliency and uniform-multiplier controls.
8. Optional adaptive-GRPO ablation, clearly separated from AD-CISPO.

Metrics. Report both task performance and optimizer behavior:

- exact-match accuracy, verifier reward, and pass@k by dataset and difficulty tier;
- reward-per-token and reward-per-GPU-hour;
- upper-clip saturation rate and token update coverage;
- entropy, approximate KL to the reference model when logged, and response length;
- AD-CISPO multiplier mean, min, max, saliency mean, and saliency concentration;
- high-saliency token category frequencies, including comma, punctuation, numeric, mathematical-operator, and answer-token fractions;
- sink-guard removed-token fraction and removed-saliency-mass fraction;
- saliency-extractor overhead in wall-clock time and peak memory, separately for target-policy causal-tangent and frozen-reference baselines.

Decision rule. AD-CISPO should be considered successful only if it improves exact-match or reward efficiency over scalar CISPO at comparable compute, or if it provides a clear stability advantage such as lower collapse rate or smoother reward improvement. A gain that also appears for shuffled saliency should be treated as evidence for variance injection or changed clipping distribution, not for the proposed saliency mechanism.

12 Milestones

1. **Week 1:** verify MiniMax-style CISPO against GRPO on GSM8K with matched model, batch size, rollout count, and reward function.
2. **Week 2:** implement causal-tangent saliency, sink guards, and top-token category logging; run short smoke tests to verify that high-saliency tokens are not dominated by comma or boilerplate punctuation.

3. **Week 3:** run AD-CISPO causal-tangent, scalar CISPO, future-attention in-degree, KV-norm, shuffled, and uniform ablations with matched seeds.
4. **Week 4:** analyze reward curves, clipping dynamics, saliency concentration, sink-guard behavior, runtime overhead, and representative trajectory changes.

13 Risks, Ablations, and Fallbacks

Risk: attention saliency is not causal. Attention weights may not explain model behavior [8]. This is why the next-generation plan treats future-attention in-degree as a diagnostic baseline and uses causal-tangent saliency as the primary extractor. The shuffled saliency ablation remains required to verify that any gain is coming from the token-importance signal rather than from multiplier variance.

Risk: attention sinks distort saliency. Initial and special tokens can attract attention for non-semantic reasons [9]. The implementation must mask prompt and sink tokens, use generated-token-only normalization, and report saliency concentration diagnostics. For causal-tangent saliency, sink guards should be kept explicit rather than hidden inside the saliency formula so that the removed token and removed-mass fractions can be audited.

Risk: causal-tangent overhead dominates. Causal-tangent saliency adds an extra objective-specific backward pass. The fallback is to compute it every N rollout batches, use only the final two decoder layers, reduce the saliency minibatch size, or fall back to KV-norm or future-attention in-degree for cheaper diagnostic runs.

Risk: budget concentration increases variance. Large multipliers may overemphasize a small set of tokens. The fallback is a bounded multiplier projection with post-clamp renormalization, plus reporting clip-min, clip-max, and saturation metrics.

14 Expected Impact

If successful, AD-CISPO would provide evidence that reasoning-RL trust regions should be token-budgeted rather than uniformly scalar. The contribution would be useful even if the exact attention heuristic is later replaced: the broader idea is to preserve the average IS clipping budget while assigning update freedom where the current model’s objective-sensitive token representations suggest the update can matter most. A negative result would still be informative because it would test whether CISPO’s scalar clipping is already sufficient, and whether token-level saliency signals are too weak or too noisy for optimizer control.

References

- [1] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal Policy Optimization Algorithms. *arXiv:1707.06347*, 2017. <https://arxiv.org/abs/1707.06347>.
- [2] Long Ouyang et al. Training language models to follow instructions with human feedback. *arXiv:2203.02155*, 2022. <https://arxiv.org/abs/2203.02155>.

- [3] Zhihong Shao et al. DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models. *arXiv:2402.03300*, 2024. <https://arxiv.org/abs/2402.03300>.
- [4] DeepSeek-AI. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. *arXiv:2501.12948*, 2025. <https://arxiv.org/abs/2501.12948>.
- [5] Qiying Yu et al. DAPO: An Open-Source LLM Reinforcement Learning System at Scale. *arXiv:2503.14476*, 2025. <https://arxiv.org/abs/2503.14476>.
- [6] MiniMax et al. MiniMax-M1: Scaling Test-Time Compute Efficiently with Lightning Attention. *arXiv:2506.13585*, 2025. <https://arxiv.org/abs/2506.13585>.
- [7] Ashish Vaswani et al. Attention Is All You Need. *arXiv:1706.03762*, 2017. <https://arxiv.org/abs/1706.03762>.
- [8] Sarthak Jain and Byron C. Wallace. Attention is not Explanation. *arXiv:1902.10186*, 2019. <https://arxiv.org/abs/1902.10186>.
- [9] Guangxuan Xiao et al. Efficient Streaming Language Models with Attention Sinks. *arXiv:2309.17453*, 2024. <https://arxiv.org/abs/2309.17453>.
- [10] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and Editing Factual Associations in GPT. *arXiv:2202.05262*, 2022. <https://arxiv.org/abs/2202.05262>.
- [11] Kevin Wang et al. Interpretability in the Wild: A Circuit for Indirect Object Identification in GPT-2 Small. *arXiv:2211.00593*, 2022. <https://arxiv.org/abs/2211.00593>.
- [12] Aaquib Syed et al. Attribution Patching Outperforms Automated Circuit Discovery. *arXiv:2310.10348*, 2023. <https://arxiv.org/abs/2310.10348>.
- [13] Karl Cobbe et al. Training Verifiers to Solve Math Word Problems. *arXiv:2110.14168*, 2021. <https://arxiv.org/abs/2110.14168>.
- [14] Arkil Patel, Satwik Bhattamishra, and Navin Goyal. Are NLP Models really able to Solve Simple Math Word Problems? *arXiv:2103.07191*, 2021. <https://arxiv.org/abs/2103.07191>.
- [15] Dan Hendrycks et al. Measuring Mathematical Problem Solving With the MATH Dataset. *arXiv:2103.03874*, 2021. <https://arxiv.org/abs/2103.03874>.
- [16] Hunter Lightman et al. Let’s Verify Step by Step. *arXiv:2305.20050*, 2023. <https://arxiv.org/abs/2305.20050>.
- [17] Chaoqun He et al. OlympiadBench: A Challenging Benchmark for Promoting AGI with Olympiad-Level Bilingual Multimodal Scientific Problems. *arXiv:2402.14008*, 2024. <https://arxiv.org/abs/2402.14008>.